HW 8

1. Evaluate the integrals.

(1) $\int x\cos 5x\,dx$

(2) $\int (\ln x)^2\,dx$

(3) $\int e^{2\theta}\sin 3\theta\,d\theta$

(4) $\int (\arcsin x)^2\,dx$

2. (a) Prove the reduction formula

$$\int \cos^n x dx = \frac{1}{n}\cos^{n-1} x\sin x + \frac{n-1}{n}\int \cos^{n-2} x dx$$

(b) Use part (a) to evaluate $\int \cos^2 x dx$.
(c) Use parts (a) and (b) to evaluate $\int \cos^4 x dx$.

3. Evaluate the integrals.

(1) $\int \tan^2 x\,dx$

(2) $\int \tan^3 x\sec x\,dx$

(3) $\int \sin 8x\cos 5x\,dx$

(4) $\int \frac{dx}{\cos x - 1}$

4. Evaluate the integrals.

(1) $\int \frac{\sqrt{x^2-9}}{x^3}\,dx$

(2) $\int \frac{x}{\sqrt{x^2-7}}\,dx$

(3) $\int \frac{dx}{\sqrt{x^2+2x+5}}x$

(4) $\int x\sqrt{1-x^4}\,dx$

5. Evaluate the integrals.

(1) $\int \frac{10}{(x-1)(x^2+9)}\,dx$

(2) $\int \frac{x+4}{x^2+2x+5}\,dx$

(3) $\int \frac{x^3}{\sqrt[3]{x^2+1}}\,dx$

(4) $\int \frac{e^{2x}}{e^{2x}+3e^x+2}\,dx$

6. (a) Use integration by parts to show that, for any positive integer n,

$$\int \frac{dx}{(x^2+a^2)^n}dx = \frac{x}{2a^2(n-1)(x^2+a^2)^{n-1}} + \frac{2n-3}{2a^2(n-1)}\int \frac{dx}{(x^2+a^2)^{n-1}}$$

(b) Use part (a) to evaluate

$$\int \frac{dx}{(x^2+1)^2} \quad \text{and} \quad \int \frac{dx}{(x^2+1)^3}$$

7. Evaluate the integrals.

(1) $\int \sqrt{\frac{1+x}{1-x}}\,dx$

(2) $\int \theta \tan^2\theta\,d\theta$

(3) $\int x^5 e^{-x^3}\,dx$

(4) $\int \frac{1}{x\sqrt{4x+1}}\,dx$

8. Evaluate the integrals.

(1) $\int x\sqrt[3]{x+c}\,dx$

(2) $\int \sqrt{x}e^{\sqrt{x}}\,dx$

(3) $\int \frac{e^{2x}}{1+e^x}\,dx$

(4) $\int \frac{\sec x\cos 2x}{\sin x+\sec x}\,dx$

Bonus:
Evaluate $\int \sqrt{\tan x}\,dx$.